

Glitches in the Matrix:

What Visual Illusions Tell Us About the Functioning Mind

Oliver Collins and Dominique Makowski



About Me



Studying **BSc Psychology with Clinical Approaches** at Sussex

- Recently completed my Placement Year as a Research Assistant at the **Reality Bending Lab**, neuropsychology of reality perception
- **Joint-winner**: 2025 Jane Beattie Prize for the Best Overall 2nd Year Performance
- **Winner**: 2024 Lisa Rodrigues Prize for the Best Overall 1st Year Performance

Academic interests spanning perception, psychophysiology, research methods, and programming

- Developed four psychology-based Python packages: **PyChroma** (dot-based chromostereopsis illusion generator), **PyGrid** (grid illusion generator), **AfterPy** (afterimage illusion generator), and **Resp-Music** (respiration-controlled music interface)



/OliverACollins



/in/oliver-collins



oc236@sussex.ac.uk

Visual Illusions

What are Visual Illusions?

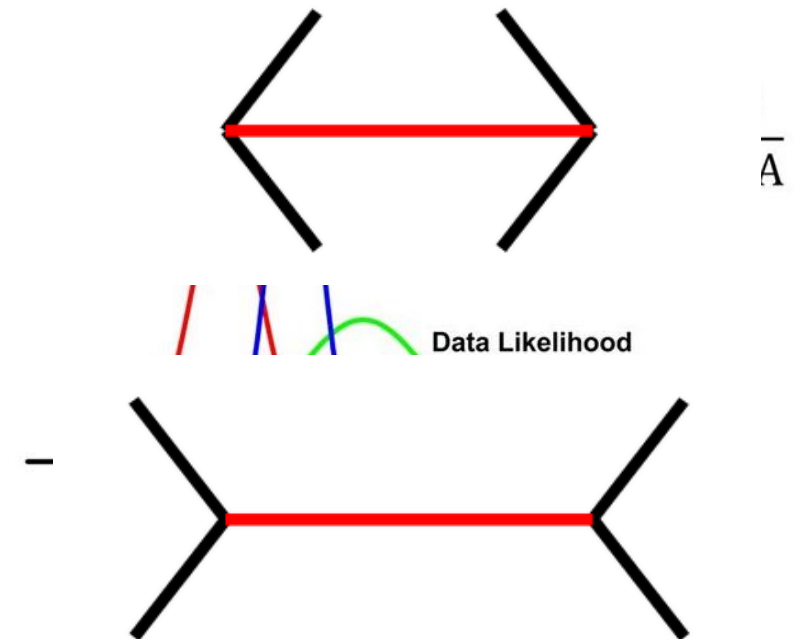
- “A **mismatch** between the objective and perceived properties of an object present in the environment” (Notredame et al., 2014)
 - **Conflict** between objective and subjective reality
 - Definitions are fuzzy
- Diversity of VI types
 - But classifications **controversial**
- Long history of interest
 - New illusions **still being created**



Hans Holbein the Younger, 'The Ambassadors', 1533

What Do VIs Tell Us?

- Canonical example of **Bayesian inference** in perception
- Elucidate fundamental mechanisms underlying the human visual perceptual system (e.g., depth perception, size constancy, conflicting cues)
- Potential interindividual differences in perception



Interindividual Differences in VI Sensitivity

Culture

- “**Carpentered-world hypothesis**” (e.g., Kroupin et al., 2025)

Psychopathology

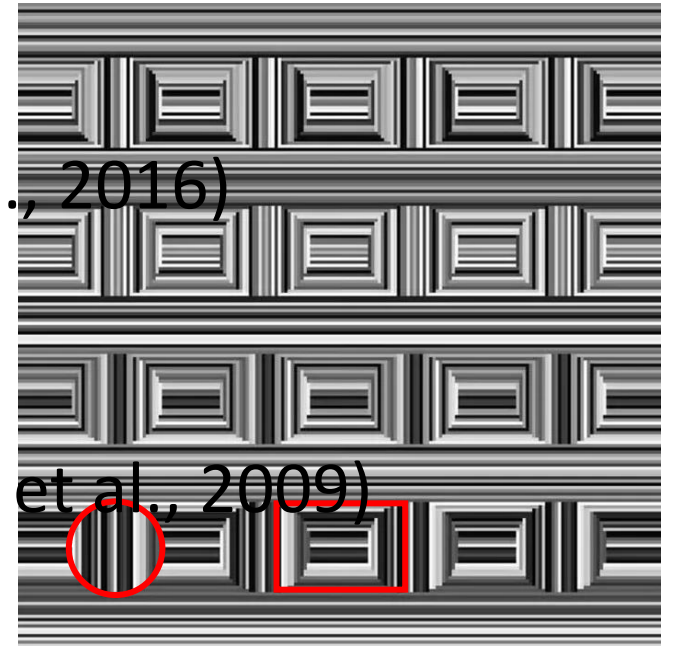
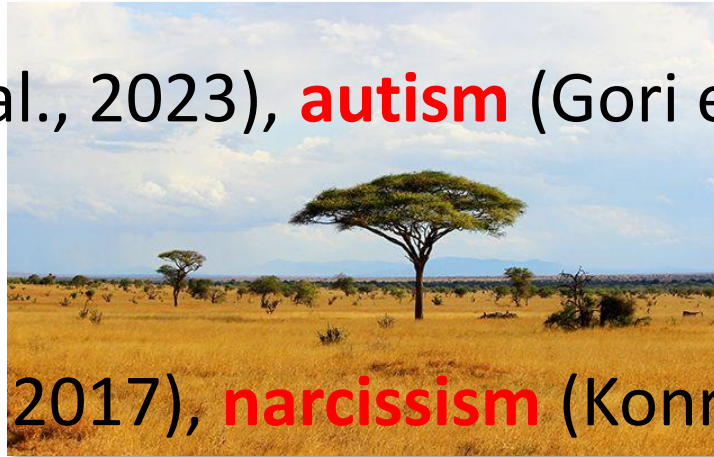
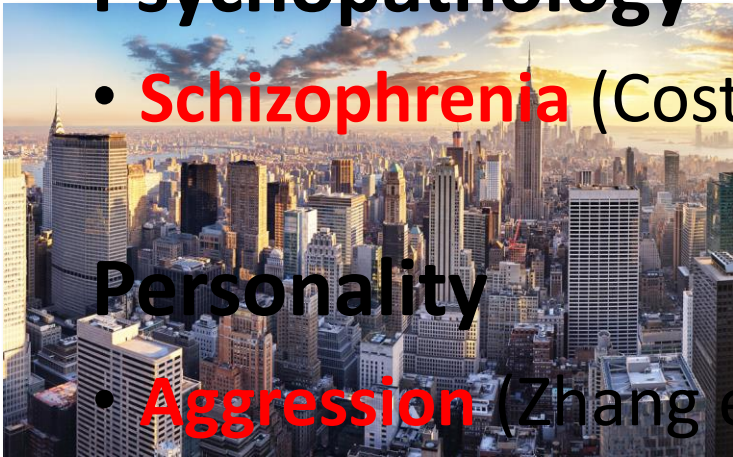
- **Schizophrenia** (Costa et al., 2023), **autism** (Gori et al., 2016)

Personality

- **Aggression** (Zhang et al., 2017), **narcissism** (Konrath et al., 2009)

VS

BUT, mixed findings...



The Coffer illusion

Approaches to Measuring VI Sensitivity

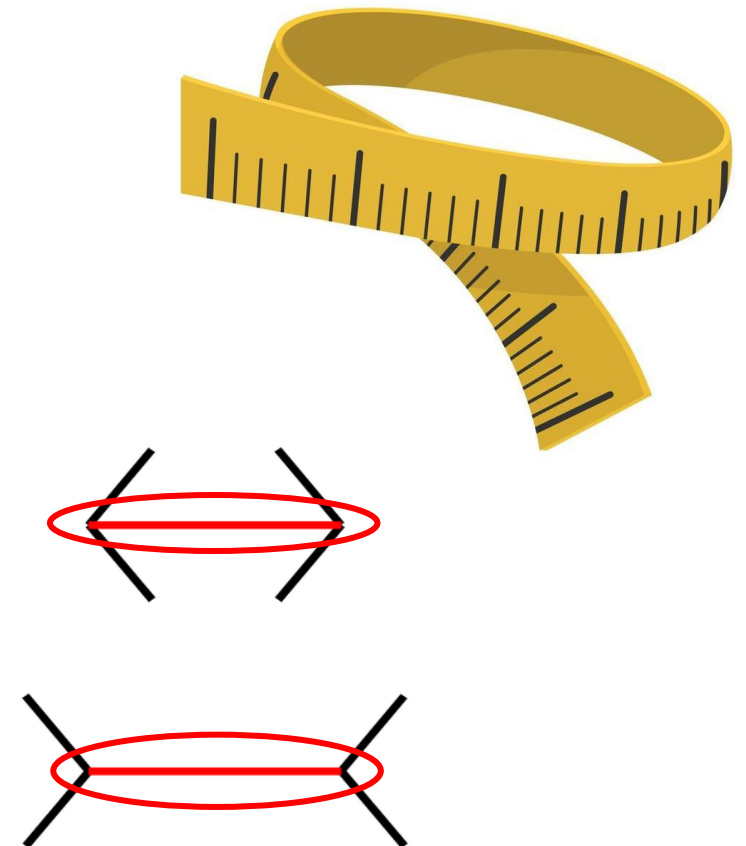
Likert and sliding scales, adjustment tasks, estimation tasks

- Reliance on metacognitive judgements

Use the arrow keys to change the size of the bottom horizontal line to match the size of the top horizontal line

Typical shortcomings of illusion paradigms

- Low number of trials
- Narrow range of illusion types
- No manipulation of the **illusion context**



How can we solve these issues?

Introducing: the Illusion Game!

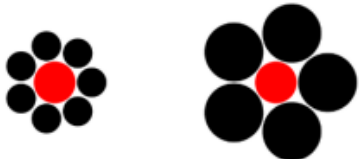
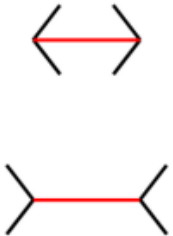
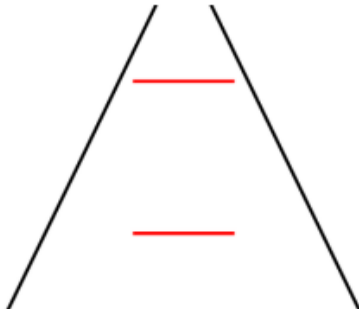
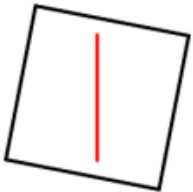
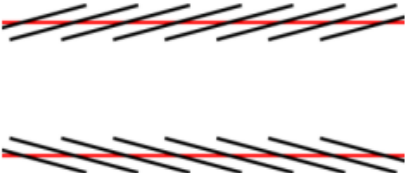
What Is the “Illusion Game”?

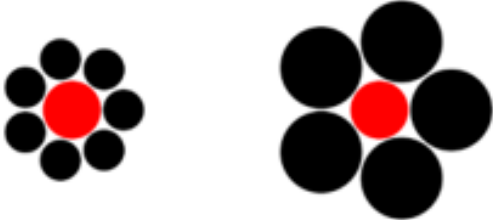
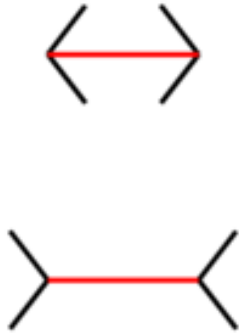
- Gamified computerised psychophysics-based 2AFC task aiming to measure **VI sensitivity** (Makowski et al., 2023)
- Participants are presented with different illusion types over 160 trials/illusion
 - Two illusion targets presented simultaneously, arrow key response made
- Illusion stimuli generated using the *Pyllusion* package (Makowski et al., 2021)
 - **Parameterised, direct manipulation** of illusions
 - Avoids influence of metacognition
- Indexes **speed-accuracy trade-off**

A Parametric Framework to Generate Visual Illusions Using Python

Dominique Makowski ¹, Zen J. Lau², Tam Pham³, W. Paul Boyce ⁴, and S.H. Annabel Chen

Py\\usion

Illusion Type	Example	Task
Ebbinghaus		<i>Which red circle is bigger?</i> 
Müller-Lyer		<i>Which red line is longer?</i> 
Vertical-Horizontal		<i>Which red line is longer?</i> 

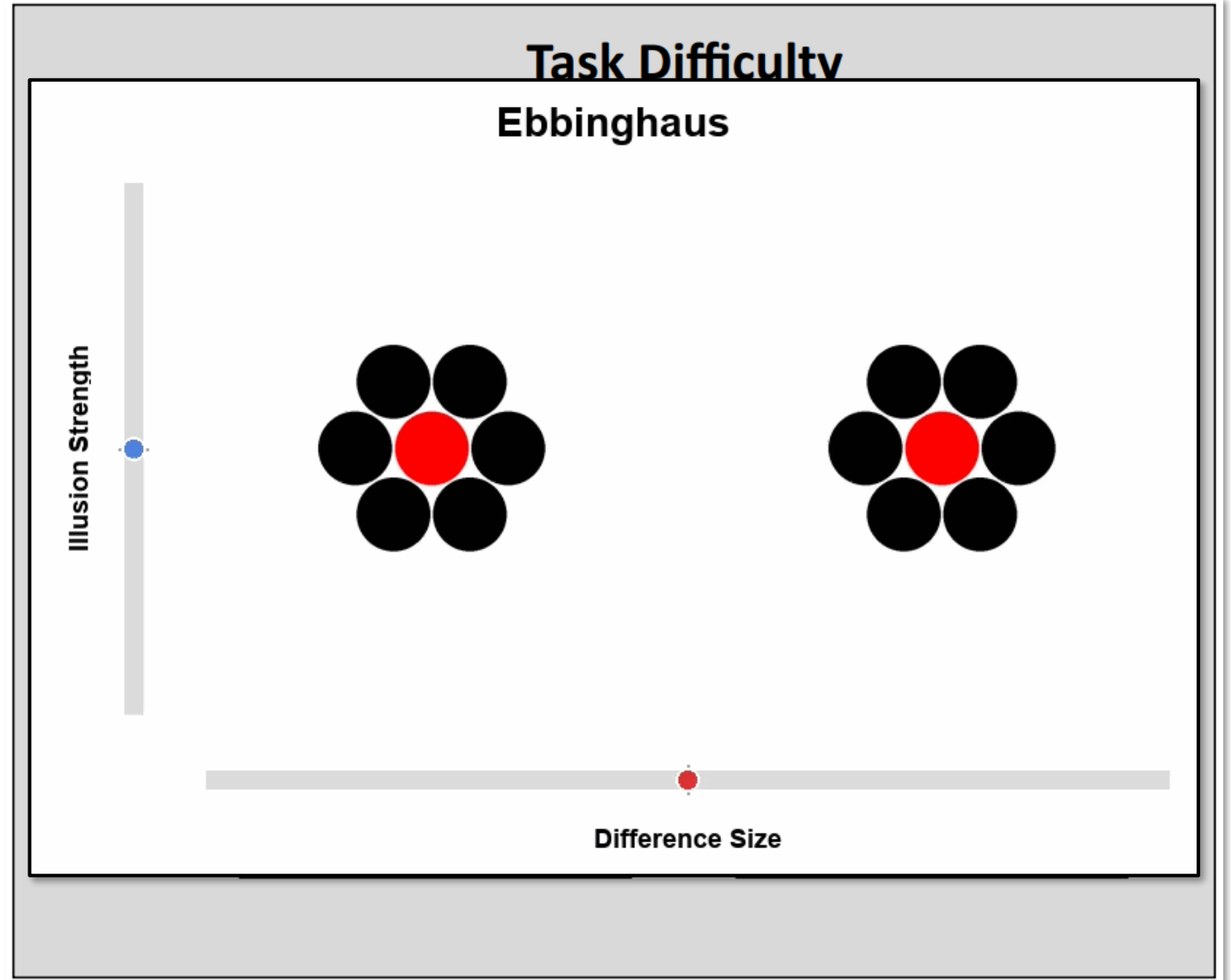
Parameters of the Illusion Game

Task Difficulty

- The relative objective size difference between **core illusion elements**

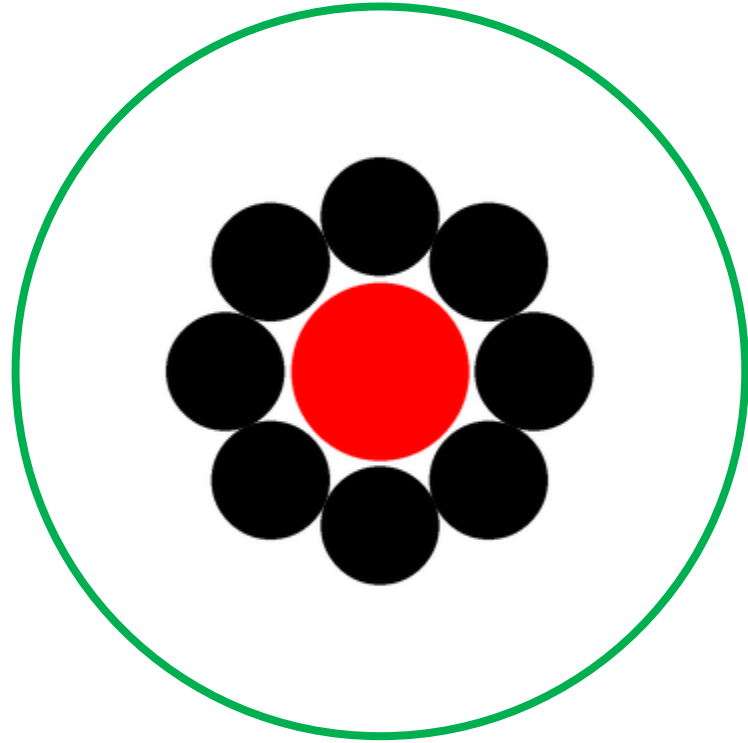
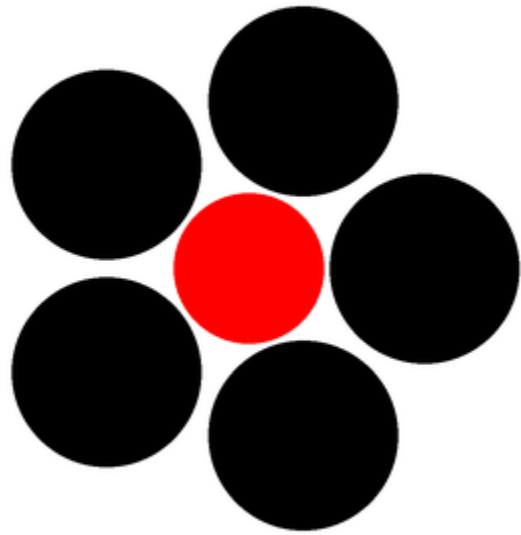
Illusion Strength

- The extent of perceptual interference from other illusion elements (**the context**)

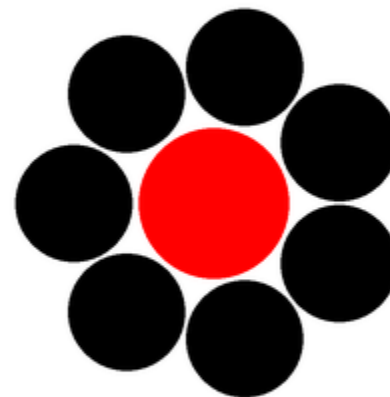
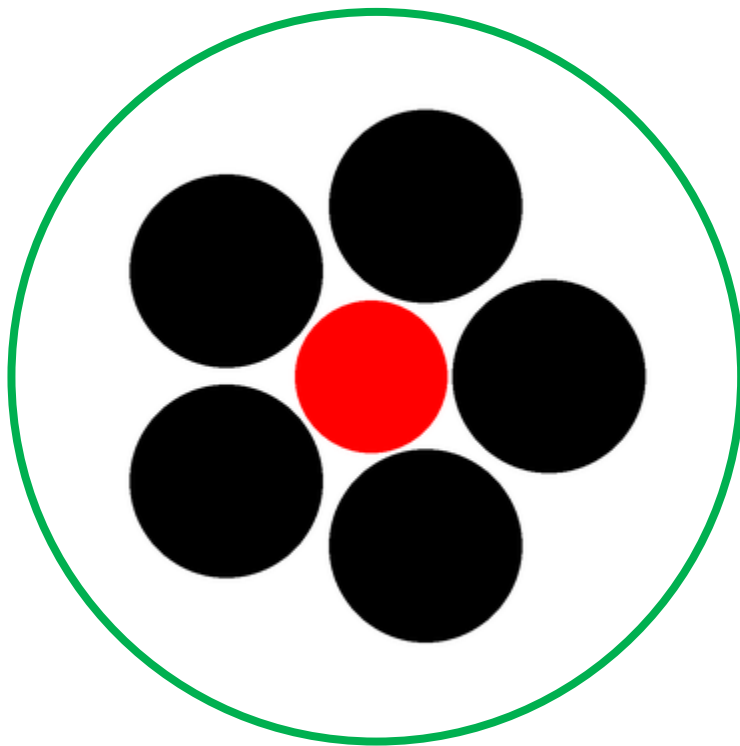


Let's play the Illusion Game!

Which red circle is bigger?



Which red circle is bigger?



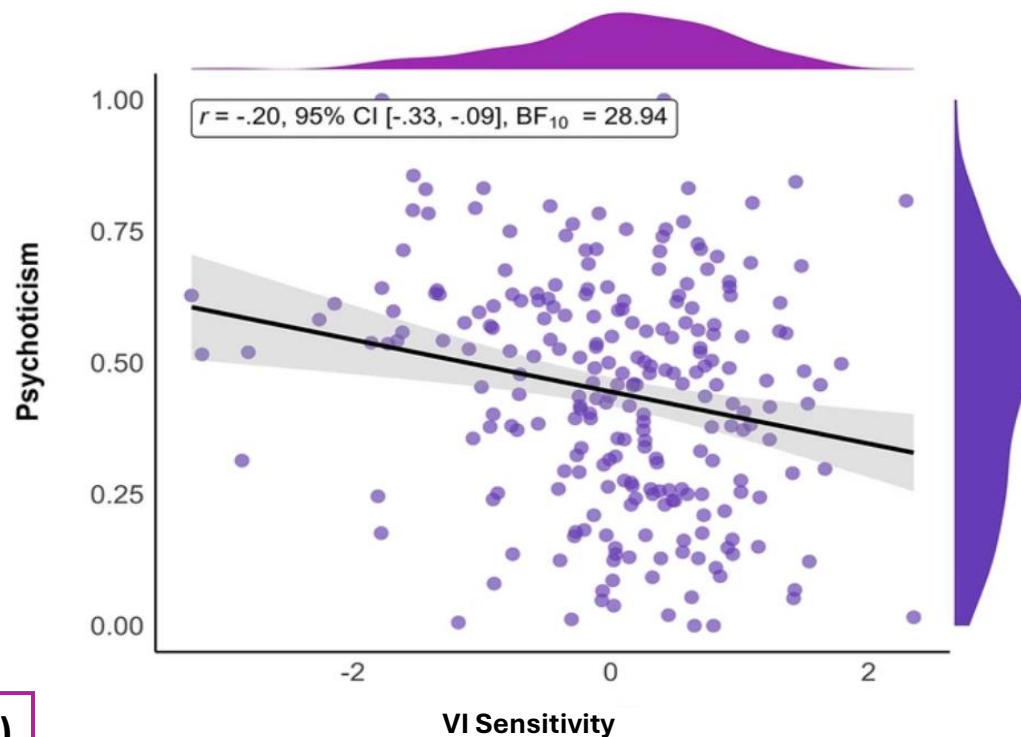
Implementing the Illusion Game

VIs and Personality Traits

Article | [Open access](#) | Published: 22 April 2023

A novel visual illusion paradigm provides evidence for a general factor of illusion sensitivity and personality correlates

[Dominique Makowski](#) , [An Shu Te](#), [Stephanie Kirk](#), [Ngoi Zi Liang](#) & [S. H. Annabel Chen](#)

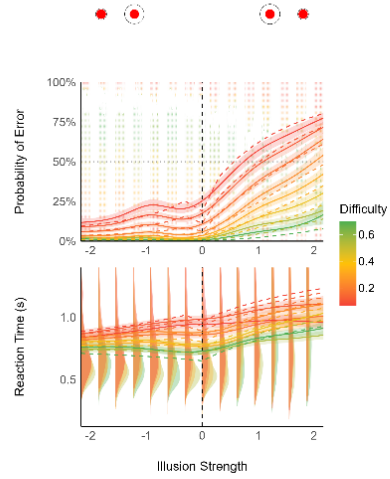


Makowski et al. (2023)

Illusion	Example	Task	Description
Delboeuf		Which red circle is bigger? 	Two circles surrounded by outer rings. The circle with the closer outline appears larger than the one with the distant outline.
Ebbinghaus		Which red circle is bigger? 	Two circles surrounded by other circles. The circle surrounded by smaller circles appears larger than the one with the larger surrounding circles.
Müller-Lyer		Which red line is longer? 	Two parallel segments that end with inwards/outwards pointing arrow-like fins. The segment with inward-pointing fins is typically perceived to be longer.
Ponzo Illusion		Which red line is longer? 	Two parallel segments embedded between a pair of converging lines. The segment on the convergence side is typically perceived to be longer.
Vertical-Horizontal		Which red line is longer? 	Two lines segments, one horizontal and one angled. The angled line is usually perceived as longer.
Zöllner		Which direction are the red lines converging towards? 	Represented by two long lines crossed with short, repeated lines, that appear as converging to one side.
Rod and Frame		Which direction is the red line leaning towards? 	A segment ("rod") enclosed in a tilted square ("frame"), that appears to be leaning to one side.
Poggendorff		Is the right line above or below the left line? 	Two oblique segments separated by a rectangle. The right segment appears to be vertically misplaced.
Simultaneous Contrast		Which small rectangle is lighter? 	Consists of two smaller grey rectangles embedded in backgrounds of different contrasts. The rectangle with the darker background appears lighter.
White		Which vertical rectangle is lighter? 	Composed of a series of contrasting horizontal bars superimposed with vertical grey rectangles. Rectangles superimposed onto darker bars appear lighter.

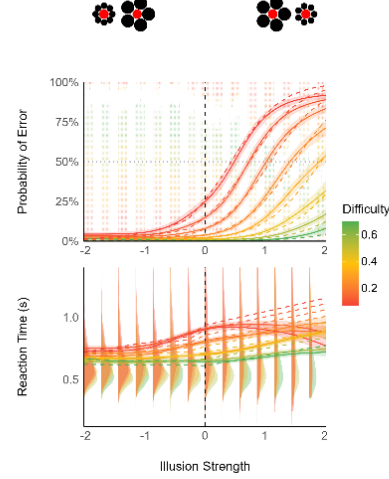
Delboeuf Illusion

Which red circle is bigger?



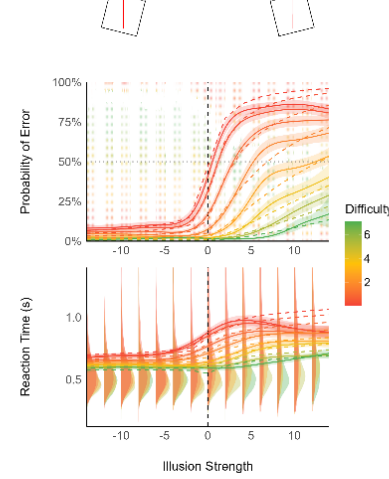
Ebbinghaus Illusion

Which red circle is bigger?



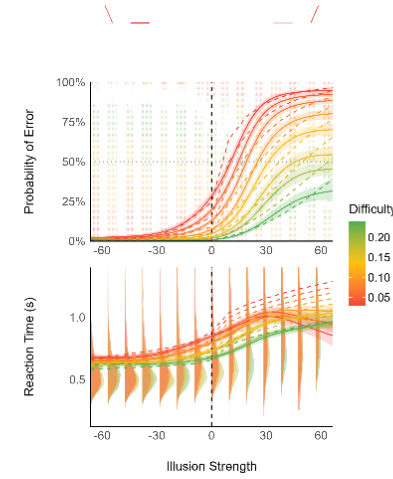
Rod-Frame Illusion

Which direction is the red line leaning towards?



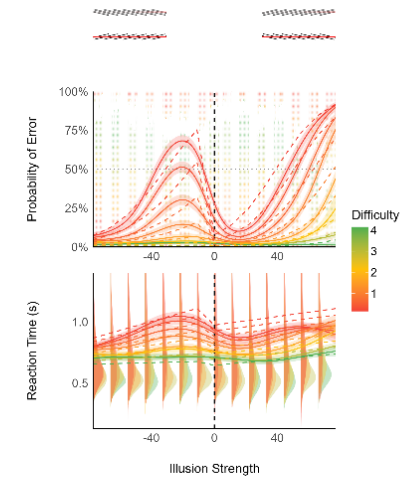
Vertical-Horizontal Illusion

Which red line is longer?



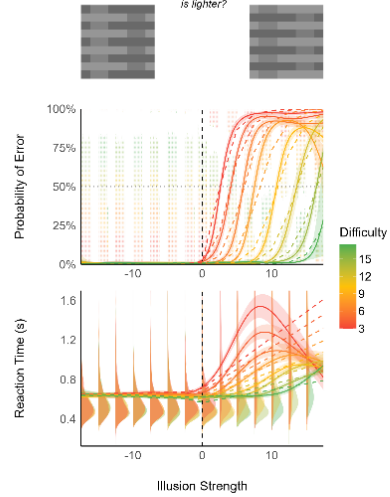
Zöllner Illusion

Which direction are the red lines converging towards?



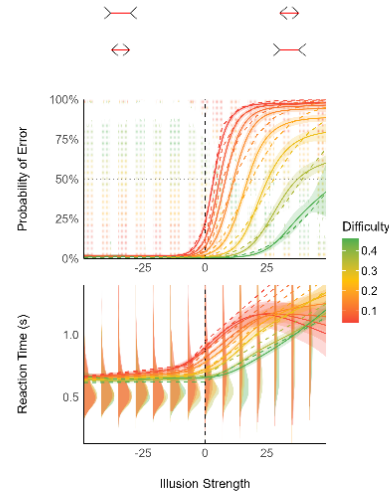
White Illusion

Which vertical rectangle is lighter?



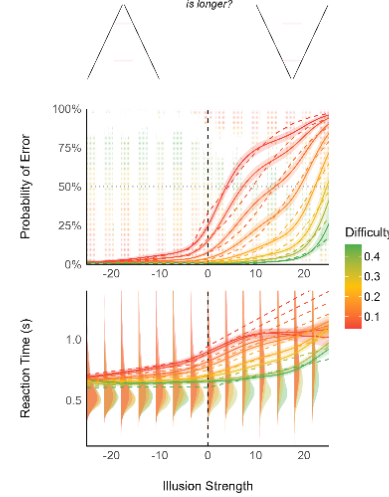
Müller-Lyer Illusion

Which red line is longer?



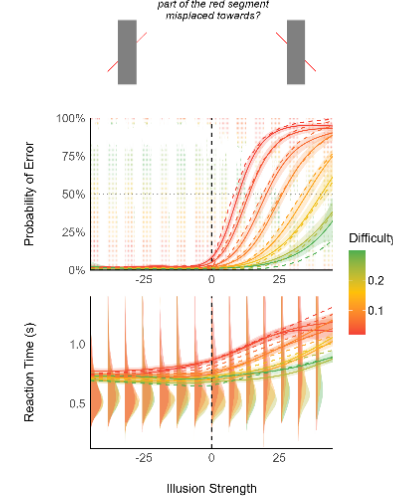
Ponzo Illusion

Which red line is longer?



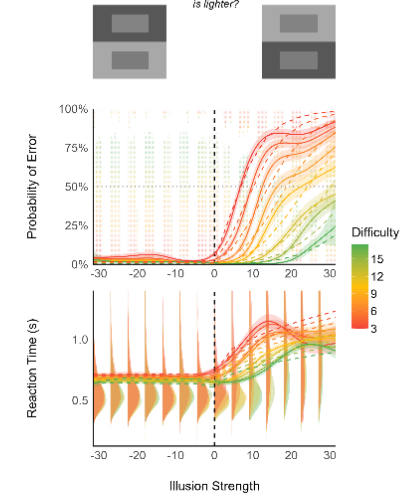
Poggendorff Illusion

Which side is the right part of the red segment misplaced towards?



Contrast Illusion

Which small rectangle is lighter?



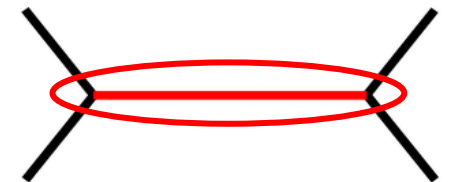
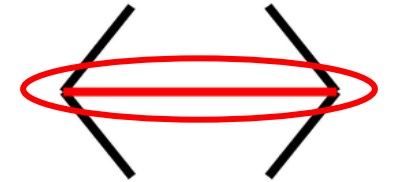
VIs and Phenomenological Control

Phenomenological Control

- Lush et al. (2021): “the ability to generate experiences to meet expectancies”
- Is VI sensitivity related to one’s ability to “**resist**” an illusion stimulus?

Lush et al. (2022)

- Found an **absence** of relationship between PC and VI sensitivity
- **Cognitive impenetrability hypothesis** (Pylyshyn, 1999)



Does this finding replicate when using the Illusion Game?

VIs and Phenomenological Control (Cont.)

What did we do?

Tested the relationship between:

1. **VI sensitivity and PC** (Lush et al., 2022)
2. **VI sensitivity and psychoticism** (Makowski et al., 2023)

Testing the Relationship between Phenomenological Control related to Illusion Sensitivity

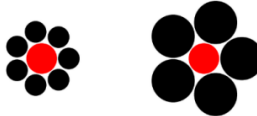
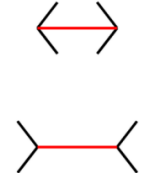
Dominique Makowski and Ana Neves

School of Psychology, University of Sussex

$$\text{IES} = \frac{\text{Mean RT of correct responses}}{\text{Proportion of correct responses}}$$

What did we find?

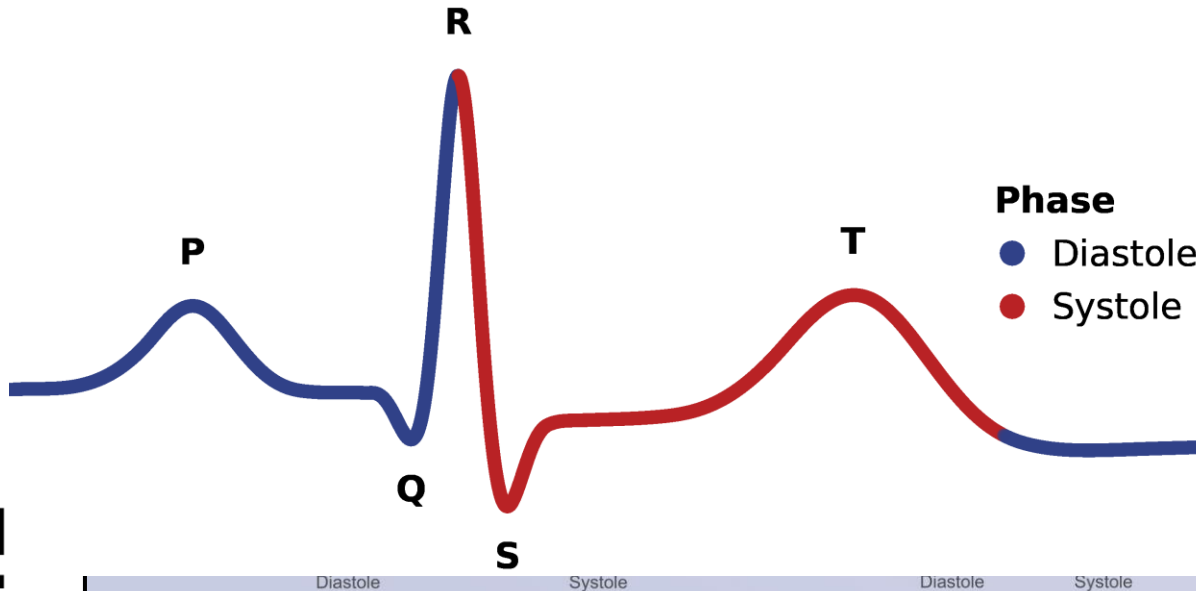
- **Absence** of relationship for both!

Illusion Type	Example	Task
Ebbinghaus		Which red circle is bigger? 
Müller-Lyer		Which red line is longer? 
Vertical-Horizontal		Which red line is longer? 

Present and Future Directions

Physiological Correlates of VI Sensitivity

- H_1 : VI sensitivity will be stronger when viewing VIs during **systole** (relative to **diastole**)



Statistical modelling (logistic regression) found **no** evidence that cardiac phase influenced **error rate**, either overall, $\chi^2(1) = 1.89, p = .169$, or in interaction with illusion type, $\chi^2(2) = 3.07, p = .216$, difficulty, $\chi^2(1) = 2.56, p = .109$, or the three-way combination of phase, illusion type, and difficulty, $\chi^2(2) = 3.81, p = .149$

ROLE OF THE CARDIAC CYCLE IN THE PERCEPTION OF REALITY AND ILLUSION

Oliver Collins¹ and Dominique Makowski^{1,2}

¹School of Psychology, University of Sussex; ²Sussex Centre for Consciousness Science, University of Sussex



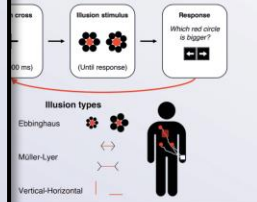
AIMS AND OBJECTIVES

To investigate whether the cardiac cycle is related to the perception of VIs

Hypothesis: In the "Illusion Game" task, VI sensitivity will be stronger (more error-prone) when participants view VIs during **systole** (relative to **diastole**)

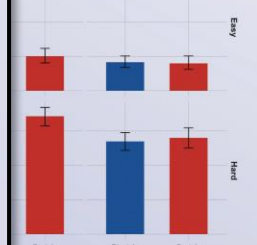
PARTICIPANTS

Participants ($M_{age} = 25.2, SD = 8.7$, range: [18, 66], 50% ♀) were recruited at the University of Sussex



Difficulty

$\chi^2 = 2.56, p = .109$ | 3-way: n.s. ($\chi^2 = 3.81, p = .149$)



Overall, $\chi^2(1) = 1.89, p = .169$, or in interaction with illusion type, and difficulty, $\chi^2(2) = 3.81, p = .149$

B. FUTURE DIRECTIONS

- We aim to carry out the following:
- Recruit 100 total participants
- Develop more complex computational models to simulate decision processes underlying VIs
- Include measures pertaining to interoception, cognitive control, and cardiac cycle % completed

Present and Future Directions (Cont.)

Computational Cognitive Modelling

- Employing **evidence accumulation models** to IG data
- Ideal for modelling speed-accuracy data

cogmod

documentation cogmod cran v0.3.3 R-CMD-check passing models list downloads null

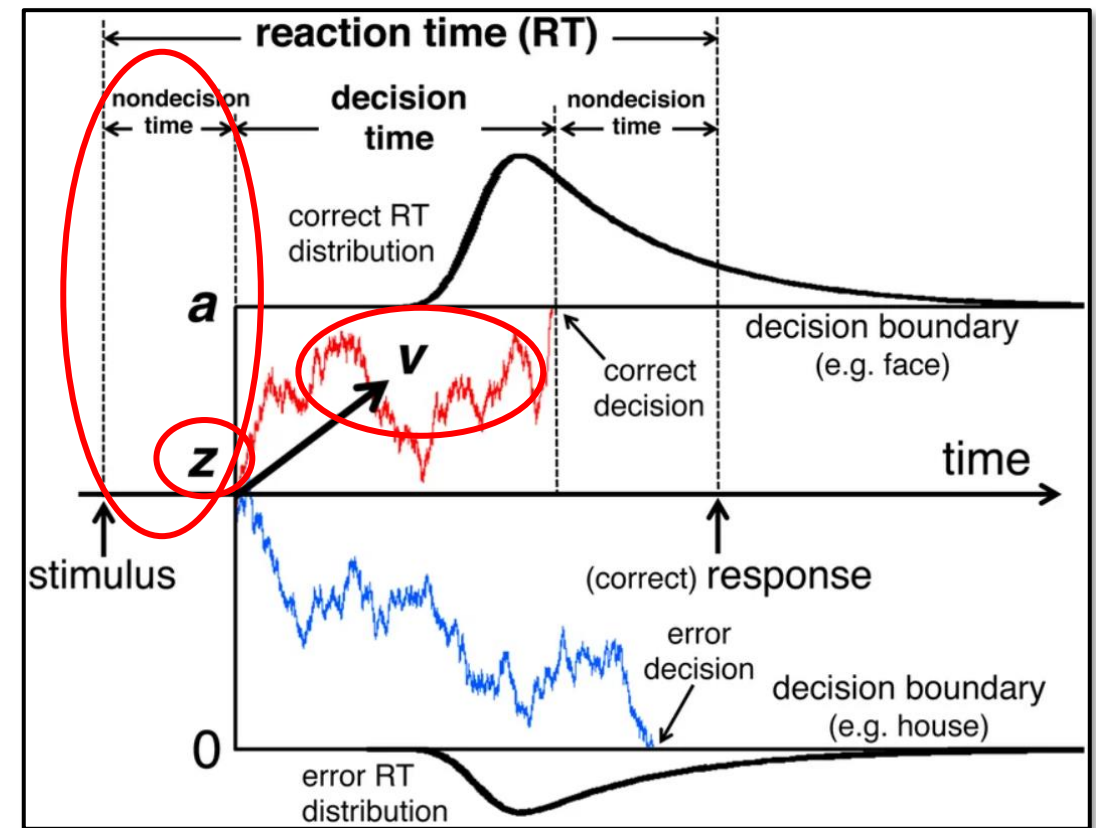
Models of Cognition for Subjective Scales and Decision Making Tasks in R

This R package is dedicated to facilitate the application of computational models of cognition in R under a Bayesian framework. These are useful in the field of cognitive science and computational neuropsychology.

If you have suggestions for improvement, please [get in touch!](#)



github.com/DominiqueMakowski/cogmod



Drift-diffusion modelling

Thank you for listening!



/OliverACollins



/in/oliver-collins



oc236@sussex.ac.uk